



MASENO UNIVERSITY
UNIVERSITY EXAMINATIONS 2017/2018

**SECOND YEAR SECOND SEMESTER EXAMINATION FOR
THE DEGREE OF BACHELOR OF SCIENCE IN CHEMISTRY WITH
INFORMATION TECHNOLOGY**

MAIN CAMPUS

SCH 205: INORGANIC CHEMISTRY II

Date: 9th July, 2018

Time: 8.30 - 11.30am

INSTRUCTIONS:

- Answer ALL Questions in Section A and any other FOUR in Section B.



SCH 205: INORGANIC CHEMISTRY II

INSTRUCTIONS:

TIME: 3 HOURS

Answer question ONE in section A and any other FOUR questions in section B

SECTION A 30 Marks (COMPULSORY)

QUESTION ONE

- a) Confirm that the experimentally observed electronic configuration of K, $1s^2 2s^2 2p^6 3s^2 3p^6 4s^1$, is energetically more stable than the configuration $1s^2 2s^2 2p^6 3s^2 3p^6 3d^1$.

[3 marks]

- b) Below is a sketch of modern Periodic Table. Use it to answer the questions below:

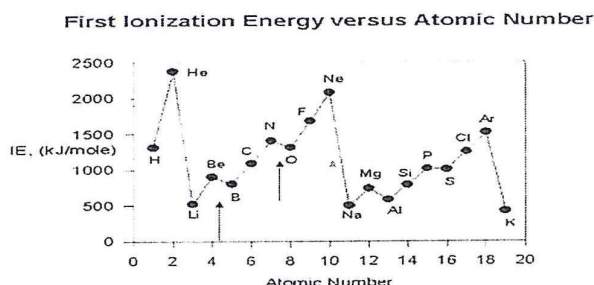
X																
Y																

- (i) Partition the periodic Table into its various blocks of elements. [2 marks]

- (ii) What general name is given to:

I.	Group 15 of the periodic Table	[1 mark]
II.	Group 16 of the Periodic Table	[1 mark]
III.	Period marked X	[1 mark]
IV.	Period marked Y	[1 mark]

- c) (i) What is the expected bond length for a single Sn-O bond?
 Given that for O, $r_{\text{cov}} = 0.81 \text{ \AA}$ and for Sn, $r_{\text{cov}} = 1.61 \text{ \AA}$ [1 mark]
- (ii) Explain how differences in electronegativities of the bonding atoms affect the above (3b) prediction in bond length. [2 marks]
- d) The graph below represents changes in ionization potentials of some elements in the Periodic Table.



Explain the anomalies indicated by the arrows between the following:

- (i) Be and B, [2 marks]
- (ii) N and O, [1 mark]
- (iii) Ne and Na. [1 mark]
- e) (i) Define the term electronegativity. [1 mark]
- (ii) Given that $D_{\text{H-H}}$ and $D_{\text{F-F}}$ in H_2 and F_2 are 436 and 158 kJ mol^{-1} , and that the bond dissociation enthalpy of HF is 297 kJ mol^{-1} . Determine electronegativity of F using Pauling method, if electronegativity of H is 2.2. [3 marks]
- $$D_{\text{H-F}} = \sqrt{(D_{\text{H-H}})(D_{\text{F-F}})} + 1.3(\chi_{\text{F}} - \chi_{\text{H}})^2$$
- f) Explain why F has a lower electron affinity than Cl. [2 marks]
- g) Comment on the reactions of dihydrogen with:
- (i) chlorine, [1 mark]
- (ii) sodium, [1 mark]
- (iii) copper(II) oxide [1 mark]
- h) (i) Would you expect the hydrides of N, O and F to have lower boiling points than the hydrides of their subsequent group members? Give reasons. [3 marks]
- (ii) Can phosphorus with outer electronic configuration $3s^2 3p^3$ form PH_5 ? [2 marks]

SECTION B 40 Marks

QUESTION TWO

- (a) KMnO_4 does not dissolve in CHCl_3 , but on addition of small amounts of crown ether, it readily dissolves. Explain. [2 marks]
- (b) Alkali metals dissolve in liquid ammonia forming blue dilute solutions which are conductive and paramagnetic. Use an equation to explain these properties of the solution. [3 marks]
- (c) Explain the origin of flame colour observed when alkali metals are put in a flame. [2 marks]
- (d) Because of 'diagonal relationship' Be behaves a lot like Al. Give three properties of Be similar to Al. [2 marks]

QUESTION THREE

- (a) State any four differences between metals and nonmetals. [4 marks]
- (b) Use a relevant chemical equation to show how chromium can be obtained using the Goldschmidt process. [2 marks]
- (c) Arrange the following in order of increasing atomic radius: Al, C, Si. [2 marks]
- (d) Using a periodic table only, arrange the following elements in order of increasing ionization energy: Ar, Se, S. [2 marks]

QUESTION FOUR

- (a) Complete the following reactions: [2 marks]
 - i) $\text{Li}_2\text{O}(\text{s}) + \text{H}_2\text{O}(\text{l})$
 - (iii) $\text{KO}_2(\text{s}) + \text{H}_2\text{O}(\text{l})$
- (b) State the periodic law. [1 mark]
- (c) Oxygen, like other second-period elements, is somewhat different from the other elements in its group. List some of these differences. [3 marks]
- (d) Discuss the bonding in each of the following molecules or ions. What is the expected molecular geometry? [4 marks]
 - (i) Cl_2O
 - (ii) BrO_3^-

QUESTION FIVE

- (a) The following solid substances are in separate but unlabeled test tubes: $\text{Al}_2(\text{SO}_4)_3 \cdot 18\text{H}_2\text{O}$, $\text{BaCl}_2 \cdot 2\text{H}_2\text{O}$, KOH . Describe how you could identify the compounds by chemical tests using only these substances and water. [2 marks]
- (b) Aluminum hydroxide is an amphoteric substance. What does this mean? Write equations to illustrate. [2 marks]
- (c) Using periodic trends, arrange the following elements by increasing ionization energy: Ar, Na, Cl, Al. [2 marks]
- (d) Although phosphorus pentabromide exists in the vapor phase as PBr_5 molecules, in the solid phase the substance is ionic and has the structure $[\text{PBr}_4^+]\text{Br}^-$. What is the expected geometry of PBr_4^+ ? Describe the bonding to phosphorus in PBr_4^+ [2 marks]
- (e) Xenon tetrafluoride, XeF_4 , is a colourless solid. Give the Lewis formula for the XeF_4 molecule. What is the hybridization of the xenon atom in this compound? What geometry is predicted by the VSEPR model for this molecule? [2 marks]

QUESTION SIX

- a) Chlorine can be prepared by oxidizing chloride ion (from hydrochloric acid) with potassium dichromate, $\text{K}_2\text{Cr}_2\text{O}_7$, which is reduced to Cr^{3+} . Write the balanced equation for the reaction. [2 marks]
- b) Magnesium nitride, reacts with water to produce magnesium hydroxide and ammonia. How many grams of ammonia can you obtain from 7.50 g of magnesium nitride? [4 marks]
- c) Write balanced equations for each of the following. [2 marks]
- Bromine reacts with aqueous sodium hydroxide to give hypobromite and bromide ions.
 - Hydrogen bromide gas forms when sodium bromide is heated with phosphoric acid.
- d) Two elements are in the same group, one following the other. One is a metalloid; the other is a metal. Both form oxides of the formula RO_2 . The first is acidic; the next is amphoteric. Identify the two elements. [2 marks]



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DEGREE OF BACHELOR OF SCIENCE AND BACHELOR OF
EDUCATION SCIENCE WITH INFORMATION TECHNOLOGY**

MAIN CAMPUS

SCH 206: ORGANIC CHEMISTRY II

Date: 3rd July, 2018

Time: 3.30- 6.30 pm

INSTRUCTIONS

- Attempt ANY FIVE questions

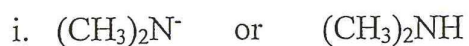


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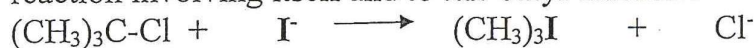
Attempt any five (5) Questions

Question one

A. Which reactant species in each of the following pairs is more reactive as a nucleophile? Explain your answer



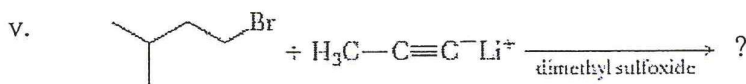
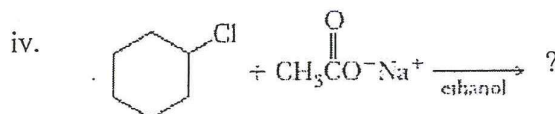
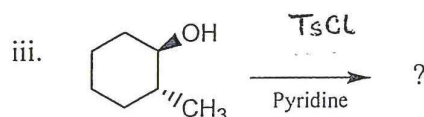
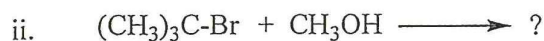
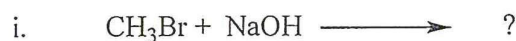
B. What would be the impact of increasing the concentration of iodide ion in a reaction involving itself and to tert-butyl chloride?



[2 marks]

C. Complete the following nucleophilic substitution reactions.

[5 marks]



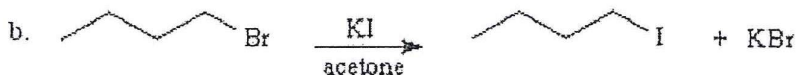
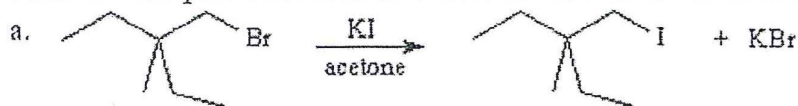
D. Suggest a suitable mechanism and write the rate law equation for the reaction in Question 1C (ii) above.

[3 marks]

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Question two

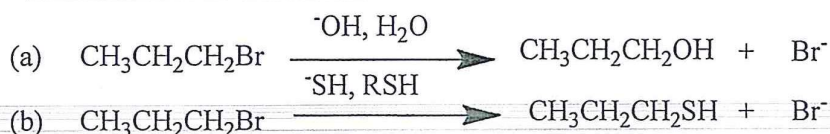
A. Consider the pair of reactions below to answer the following question(s).



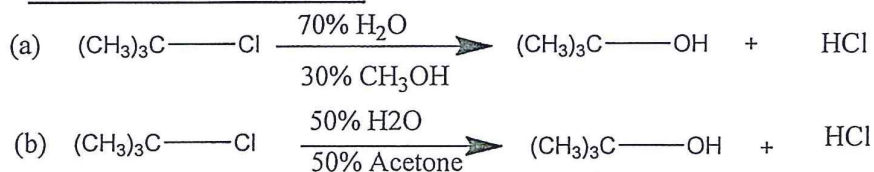
- Classify the substrate in these reactions. [1 mark]
- Classify the solvent in these reactions. [1 mark]
- Classify the mechanism for these reactions. [1 mark]
- Identify the nucleophile in these reactions. [1 mark]

B. Consider the following pairs of reaction schemes below and answer the question(s) that following:

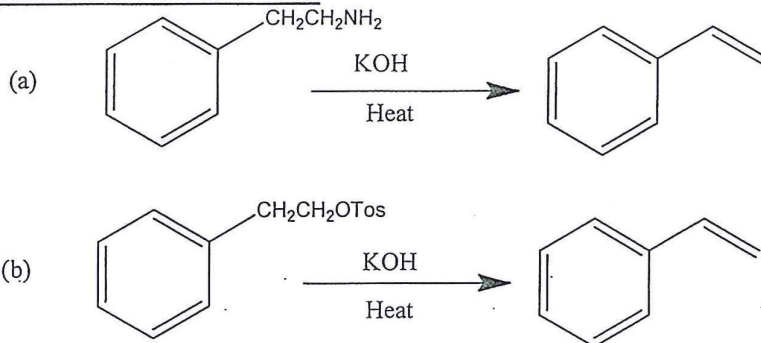
Reactions scheme set X



Reactions scheme set Y



Reaction Scheme set Z

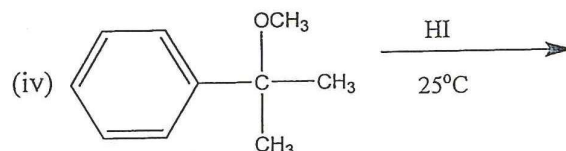
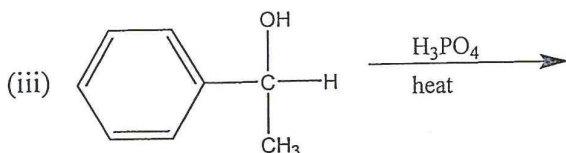
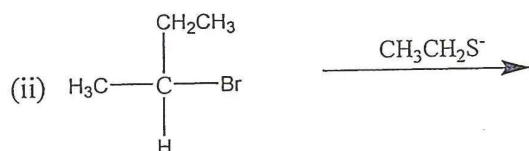
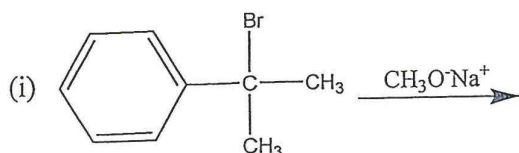


- Which reaction would you predict to be faster in each set of reaction scheme? [3 marks]
- Explain your answer to the question in part 2B (i) in each case. [7 marks]

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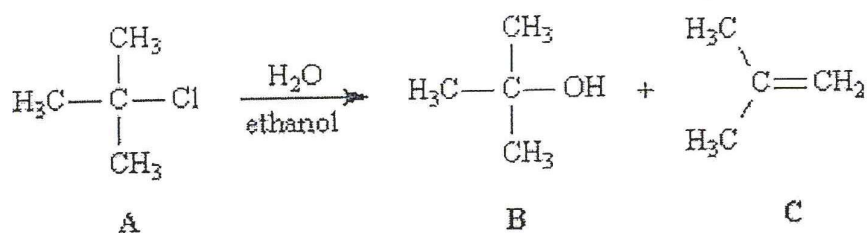
Question three

A. Indicate whether the following reactions are more likely to proceed by an E1, E2, S_N1 or S_N2 mechanism.



[4 marks]

B. Consider the reactions below to answer the following question(s).



- (i) Identify the substrate in the reaction [1 mark]
- (ii) Specify the reaction mechanism leading to Compound B and C. [2 marks]
- (iii) Write the complete reaction mechanism for the formation of Compound C in this reaction. [2 marks]

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C. To answer the following question(s) consider the narrative below:

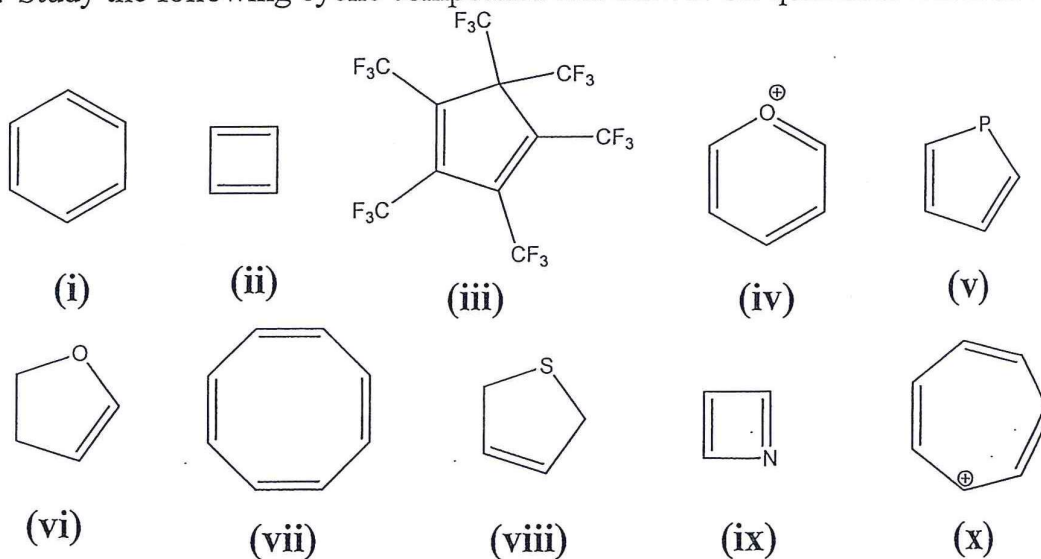
Reaction of bromomethane with sodium hydroxide in water forms methanol. If sodium iodide is added to the reaction mixture, the rate of methanol formation is dramatically increased (i.e. sodium iodide is a catalyst in this reactions).

- Write the equations of reactions involved between CH_3Br and NaOH that leads to formation of methanol. [1 mark]
- Write a reaction pathway (diagram) that accounts for the effect of added NaI and without NaI . Indicate structures for all energy minima in the diagram. [2 marks]
- Explain why adding NaI increases the reaction rate. [2 marks]

Question four

A. Aromatic compounds are defined as having planar structures, containing a certain number of electrons in a continuous overlapping cyclic arrangement of p orbitals. Name two characteristic physical or chemical properties of aromatic compounds. [3 marks]

B. Study the following cyclic compounds and answer the questions that follow.

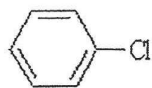


- Indicate which of the following molecules are aromatic, non-aromatic or anti-aromatic. (Assume all the molecules are planar). [10 marks]
- Briefly explain your choices for the non-aromatic compounds. [1 mark]

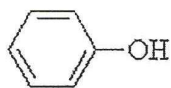
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Question five

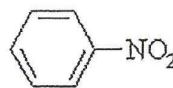
1. Arrange the following in order according to their relative rate of reaction with bromine / FeBr_3 . Justify the order.



(i)



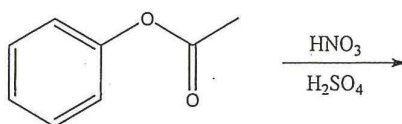
(ii)



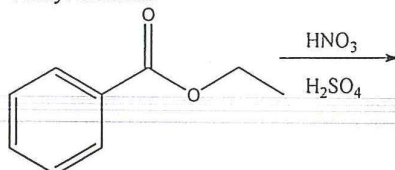
(iii)

[4 marks]

2. Draw the major products of the two reactions below and justify your answers.



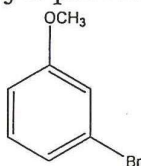
Phenyl ethanoate



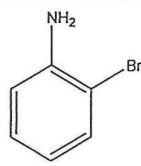
Ethyl Benzoate

[7 marks]

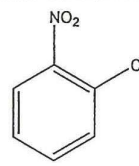
3. Draw the major products of bromination of the following compounds.



(i)



(ii)

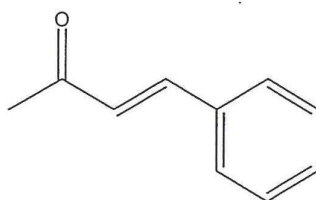


(iii)

[3 marks]

Question six

- A. Consider electrophilic aromatic substitution (EAS) on the compound *trans*-(2-methoxyethenyl)benzene.



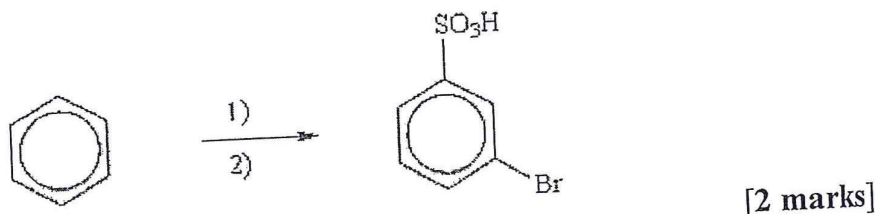
- (i) Suggest the substituent effects on the ring towards EAS. [2 mark]

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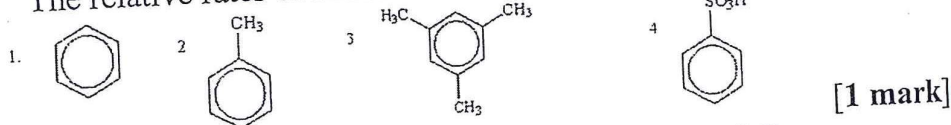
- (ii) Indicate which aromatic carbons would undergo EAS the fastest and Explain/illustrate your prediction [3 marks]

B. Provide a logical sequence in the following questions:

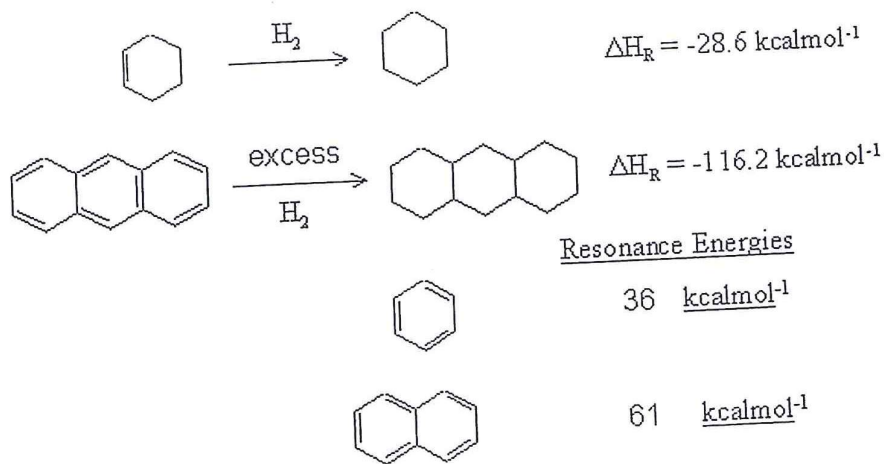
- (i) Reagent (1) and (2) in the synthetic reaction scheme below for synthesis of *m*-bromobenzenesulfonic acid from Benzene.



- (ii) The relative rates of nitration of the following compounds



C. Use the following data below to answer the question that follows.



- (i) Calculate the resonance energy of anthracene. [3 marks]
- (ii) Based on the resonance energies of polycyclic aromatics like naphthalene compared to that of benzene, suggest why they are more reactive than benzene. [3 marks]